S.C.O.U.T. · ELECTRICAL · DRAWING SET

# Electronics system: block diagram, stack-up, wiring and takeoffs

Everything that goes inside and around the buoy's electronics housing, how each part connects to the Feather M0, which cable or connector each link uses, and what is still to buy. Prototype design as of 2026-10-12. Values marked † are proposed and get confirmed on the bench.

PRELIMINARY · NOT FOR FINAL BUILD

| | |
|---|---|
| PROJECT | **S.C.O.U.T. nearshore reef buoy** |
| DRAWING NO. | SCOUT-E-100 |
| REVISION | B (draft) — audio switch to A3, 6 m cable |
| DATE | 2026-10-12 |
| SCALE | NTS |
| DRAWN | I. Rodriguez (ECEN) |
| CHECKED | — (S. Krishnan to review) |
| SHEETS | 0 Hydrophone · 1 Block diagram · 2 Stack-up · 3 Pins & cables · 4 Material takeoff · 5 Equipment takeoff · 6 Notes & references · 7 Every connection |

# Sheet 0 — Hydrophone selection: Aquarian H2dM



## Why this one

✓ **Fits the 3.3 V system:** it runs on plug-in power, at least 2 V and 0.7 mA. 3.3 V through 1.1 kΩ gives about 2.5 V. No 5 V or 48 V rail needed.

✓ **Easy to wire:** it has a standard 3.5 mm TRS plug. It goes into a jack breakout on the Proto board, so there is nothing to solder on the hydrophone itself.

✓ **Easy to read:** the output is a plain analog voltage (1.1 kΩ output impedance). A two-stage MCP6002 preamp feeds the M0's ADC on A2. No codec chip and no I²S.

✓ **Good enough for reef sound:** it covers below 10 Hz to above 100 kHz, which includes fish calls (50 Hz–3 kHz) and snapping shrimp (2–20 kHz) up to our 11 kHz sampling limit.

✓ **Rated well past our depth:** under 80 m, against a reef depth of about 1–5 m. It weighs 105 g and measures 25 × 46 mm.

✓ **Good price:** $229 base, the lowest-cost calibrated hydrophone we found that fits. The cable is 3, 6, 9 or 15 m; **we are ordering 6 m** (covers 3–6 m water depth).

Datasheet: H2dM user's guide, saved as `DATASHEETS/aquarian-h2dm-manual.pdf`. Order from aquarianaudio.com and confirm the cable-option price at checkout.

## Options compared

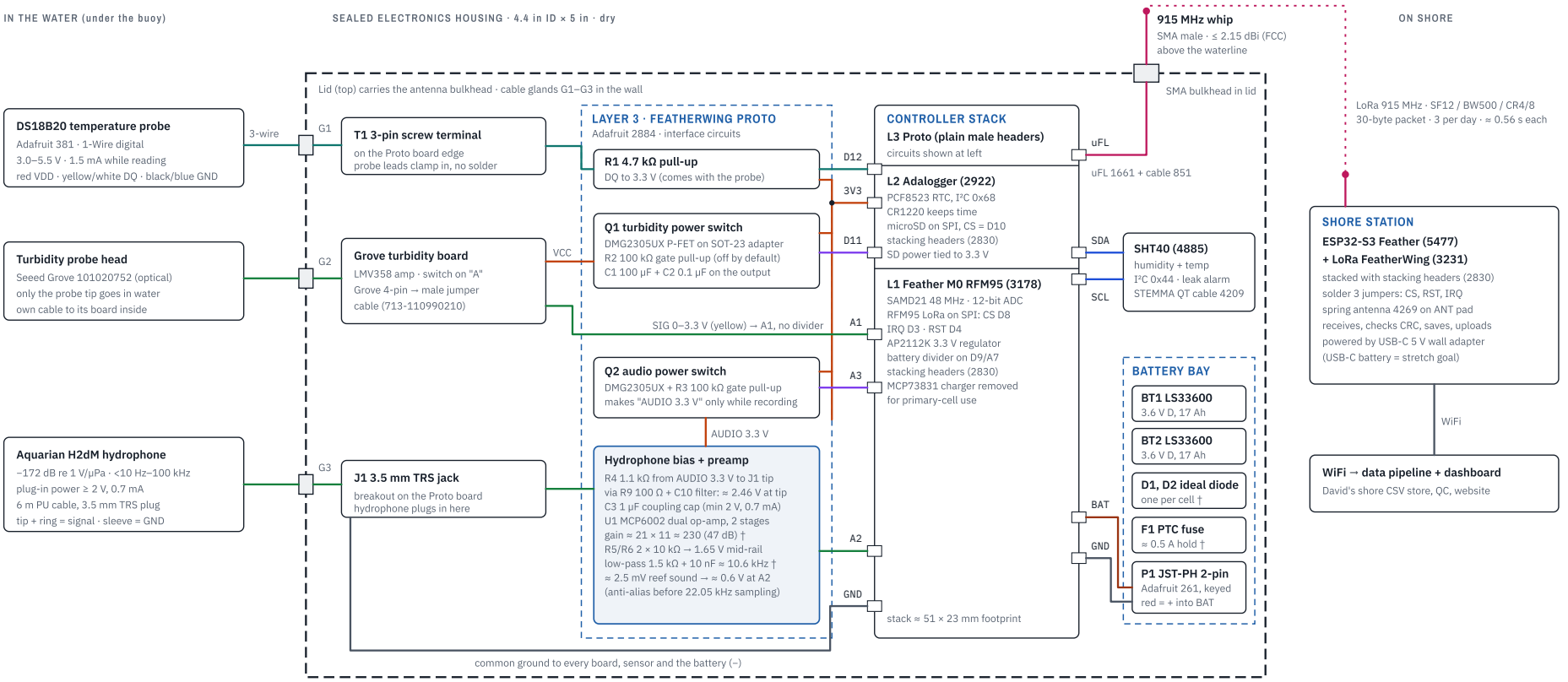
| OPTION | POWER | OUTPUT | VERDICT |
|---|---|---|---|
| **Aquarian H2dM** | Plug-in, ≥ 2 V, 0.7 mA | 3.5 mm analog | SELECTED |
| Aquarian H2dX (XLR) | Phantom power, up to 48 V | XLR balanced | Rejected: needs a high-voltage supply |
| Aquarian H2a | Plug-in | 3.5 mm | Discontinued (replaced by the H2d) |
| HydroMoth | Own battery | Records to its own SD card | Not integrated with the M0; can't schedule or summarize (unverified) |
| Aquarian A5 element | — | Bare element | Needs our own potting and preamp (unverified, about €199) |
| DIY piezo disc | — | Uncalibrated | Too cheap: no sensitivity spec, so it gives no scientific numbers |

Signal math: −172 dB re 1 V/µPa at a 120 dB re 1 µPa reef gives 10^((120−172)/20) ≈ 2.5 mV. Gain of about 230 (47 dB) brings that to about 0.6 V, centered on 1.65 V, which leaves headroom for loud events †.

# Sheet 1 — System block diagram



Battery 3.6 V · 3.3 V rail · Ground · 1-Wire data · Analog signal · I²C · GPIO on/off control · RF / LoRa




Inside the stack (no wires to add): SPI (SCK/MOSI/MISO) is shared by the LoRa radio (CS D8) and the SD card (CS D10); I²C (SDA/SCL) is shared by the clock (0x68) and the SHT40 (0x44).

Firmware wakes every 30 min on the SAMD21 internal RTC (standby), switches sensors on with D11/A3 LOW, reads, logs to SD, transmits in fixed slots, switches everything off.

Pins already used by the board: D3, D4, D8 (radio), D9/A7 (battery), D10 (SD), D13 (red LED, keep off). D5 = PCF8523 INT (firmware). Free spares: A0, A4, A5, D6 (keep free for radio IO1); D0/D1 are Serial1.

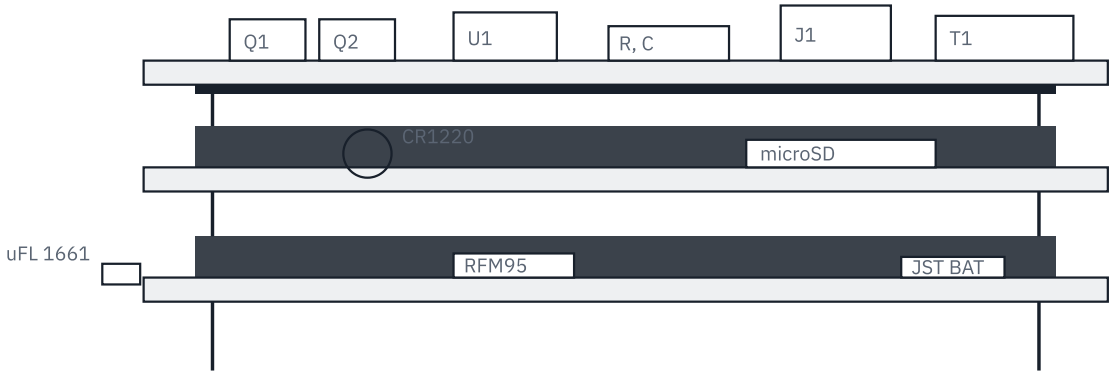
† proposed value; confirm on the bench before soldering. G1–G3: cable glands from lab stock; the 3.5 mm plug (≈ 6 mm) must pass G3 or be fitted after the gland.

# Sheet 2 — Controller stack-up and battery pack



Three Feather-format boards stack on headers: the Feather M0 at the bottom, the Adalogger in the middle, the FeatherWing Proto on top. Stacking headers (female socket on top, long pins below) let each board pass every pin up to the next. The Proto board takes plain male pins because nothing stacks on it.

SIDE VIEW (not to scale)




**L3 · FeatherWing Proto (2884)**
plain male headers down into L2 · Q1/Q2 on SOT-23 adapters, U1 MCP6002, J

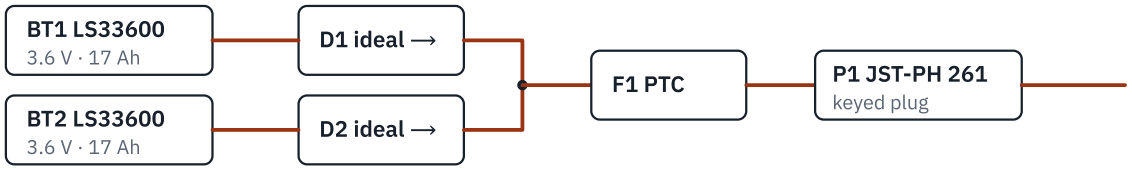
**L2 · Adalogger FeatherWing (2922)**
stacking headers (2830): sockets on top for L3, long pins down into L1

**L1 · Feather M0 RFM95 (3178)**
stacking headers (2830): long pins below go into a breadboard on the bench, s

Antenna: bench = spring 4269 on the ANT pad; deployed = uFL 1661 → cable 851 → SMA bulkhead → whip. Use one, never both.

Approximate stack height ≈ 30 mm with two stacking-header levels †. Footprint 51 × 23 mm (Feather standard).

BATTERY PACK (schematic)




**→ Feather BAT**
cells in PARALLEL (never series: 7.2 V is too high)
each cell's ideal diode stops one cell charging the other
the fuse limits a short; the keyed plug can't go in backwards

# Sheet 3 — Pin assignment and cable schedule



## Feather M0 pin assignment

| PIN | USED BY | TYPE | SOURCE OF TRUTH |
|---|---|---|---|
| D3 · D4 · D8 | RFM95 radio IRQ · RST · CS | SPI + GPIO | Feather M0 schematic |
| SCK · MOSI · MISO | Radio + SD card (shared) | SPI | Schematic; Adalogger guide p. 7 |
| D10 | SD card chip select | SPI CS | Adalogger guide p. 7 |
| SDA · SCL | PCF8523 (0x68) + SHT40 (0x44) | I²C | Adalogger p. 6; SHT40 guide |
| D9 / A7 | Battery divider (100 k / 100 k) | Analog | Schematic R3/R6 |
| D12 | DS18B20 data (DQ) | 1-Wire | Firmware config.h (decided 2026-10-12; update schematic) |
| A1 | Turbidity signal | Analog | SCO-85 |
| D11 | Q1 gate (turbidity power, LOW = on) | GPIO | This drawing |
| A3 | Q2 gate (audio power, LOW = on) | GPIO | This drawing (A3 used as digital out; D5 is taken by RTC INT) |
| A2 | Hydrophone preamp output | Analog | This drawing |
| D5 | PCF8523 INT1 wake (backup to internal RTC) | GPIO in | Firmware config.h |
| D13 | Red LED (keep off) | GPIO | Schematic |
| BAT · GND | Battery pack via JST-PH | Power | Schematic |

## Cable and connection schedule

| ID | FROM → TO | CABLE / CONNECTOR | HOW TO CONNECT |
|---|---|---|---|
| W1 | DS18B20 → T1 | Probe's own 3-core lead through G1 | Strip, lightly tin, clamp in terminal (red 3V, yellow/white D12, black/blue GND); R1 4.7 kΩ DQ→3.3 V on Proto |
| W2 | Turbidity probe → Grove board | Probe's own cable through G2 | Plugs into the Grove board; seal the probe top |
| W3 | Grove board → Proto / M0 | Grove → male jumper 110990210 | Red→Q1 out, black→GND, yellow→A1, white unused |
| W4 | H2dM → J1 | 6 m PU cable, 3.5 mm TRS through G3 | Plug into J1; tip and ring both carry signal, sleeve GND |
| W5 | SHT40 → stack | STEMMA QT → male 4209 | Red 3.3 V, black GND, blue SDA, yellow SCL |
| W6 | Battery pack → BAT | JST-PH 2-pin 261 | Red +, black –; plug only after the charger is removed |
| W7 | M0 uFL → lid | uFL 1661 + SMA cable 851 | Solder 1661 on the M0 uFL pads; 851 bulkhead nut in the lid |
| W8 | Bulkhead → whip | SMA male whip, ≤ 2.15 dBi | Hand-tighten; never transmit with nothing attached |
| W9 | Shore station power | USB-C data cable + 5 V adapter | Wall outlet at the shore site |

# Sheet 4 — Material takeoff



Every part by section. Prices are list prices per unit; ranges are estimates for parts not yet quoted. The Adafruit order (invoice 3758294) has been placed; its shipping and tax are listed once at the end.

| ITEM | PART NUMBER | DESCRIPTION | QTY | UNIT | EXTENDED | STATUS | SOURCE |
|---|---|---|---|---|---|---|---|
| **1 · Controller stack (buoy)** | | | | | | | |
| M0 | Adafruit 3178 | Feather M0 RFM95 LoRa 900 MHz | 1 | $34.95 | $34.95 | ORDERED | Adafruit |
| AD | Adafruit 2922 | Adalogger FeatherWing (RTC + microSD) | 1 | $8.95 | $8.95 | ORDERED | Adafruit |
| PR | Adafruit 2884 | FeatherWing Proto (interface circuits) | 1 | $4.95 | $4.95 | ORDERED | Adafruit |
| SH | Adafruit 2830 | Stacking headers, 12 + 16 pin (M0, Adalogger) | 2 | $1.25 | $2.50 | ORDERED | Adafruit |
| SD | Adafruit 5250 | 128 MB microSD (bench logging) | 1 | $3.95 | $3.95 | ORDERED | Adafruit |
| CC | Mouser 658-CR1220 | CR1220 3 V coin cell (RTC backup) | 1 | $1.01 | $1.01 | MOUSER | Mouser |
| **Subtotal — Controller stack (buoy)** | | | | | **$56.31** | | |
| **2 · Sensors** | | | | | | | |
| TS | Adafruit 381 | DS18B20 waterproof probe (incl. 4.7 kΩ) | 1 | $9.95 | $9.95 | ORDERED | Adafruit |
| TB | Mouser 713-101020752 | Seeed Grove turbidity sensor (3.3 V) | 1 | $20.80 | $20.80 | MOUSER | Mouser |
| TC | Mouser 713-110990210 | Grove → male jumper cable, 5-pack | 1 | $3.19 | $3.19 | MOUSER | Mouser |
| HS | Adafruit 4885 | SHT40 humidity + temperature | 1 | $5.95 | $5.95 | ORDERED | Adafruit |
| HC | Adafruit 4209 | STEMMA QT → male header cable | 1 | $0.95 | $0.95 | ORDERED | Adafruit |
| HY | Aquarian H2dM | H2dM hydrophone, 3.5 mm PIP, 6 m cable, decided 2026-10-12 (base $229 + cable option) | 1 | $229.00–249.00 | $229.00–249.00 | LATER | Aquarian Audio |
| **Subtotal — Sensors** | | | | | **$269.84–289.84** | | |

| ITEM | PART NUMBER | DESCRIPTION | QTY | UNIT | EXTENDED | STATUS | SOURCE |
|---|---|---|---|---|---|---|---|
| **3 · Interface circuits (on the Proto board)** | | | | | | | |
| Q1, Q2 | Mouser 621-DMG2305UX-7 | P-MOSFET, SOT-23 (sensor + audio power) | 2 | $0.40 | $0.80 | MOUSER | Mouser |
| AD1 | SparkFun BOB-00717 | SOT-23 to DIP adapter | 2 | $1.25 | $2.50 | MOUSER | SparkFun / Mouser |
| U1 | MCP6002-I/P | Dual rail-to-rail op-amp, DIP-8 (preamp) | 1 | $0.40–0.60 | $0.40–0.60 | LATER | DigiKey / Mouser |
| J1 | Adafruit 1699 | 3.5 mm stereo jack breakout (L = tip) | 1 | $0.95 | $0.95 | LATER | Adafruit |
| T1 | 3-pin 5 mm screw terminal | DS18B20 lead terminal | 1 | $0.50–1.00 | $0.50–1.00 | LAB | Lab |
| R1–R13 | ¼ W 1 % metal film | 4.7 k (with probe), 100 k ×3, 10 k ×3, 1 k ×2, 1.1 k, 1.5 k, 20 k, 100 Ω | 1 | $0.00–10.00 | $0.00–10.00 | LAB | Lab or resistor kit |
| C1–C10 | Capacitors | 100 µF 10 V elec.; 10 µF ×5; 1 µF ceramic; 0.1 µF ×2; 10 nF | 1 | $0.00–10.00 | $0.00–10.00 | LAB | Lab or cap kit |
| U1S | DIP-8 socket | So the op-amp can be swapped | 1 | $0.00–0.50 | $0.00–0.50 | LAB | Lab |
| H3 | Male header 12 + 16 pin | Proto → Adalogger (if the Proto ships without) | 1 | $0.00–0.50 | $0.00–0.50 | LAB | Lab / Adafruit |
| WR1 | 22 AWG solid hookup wire | Point-to-point wiring on the Proto board | 1 | $0.00–16.00 | $0.00–16.00 | LAB | Lab / Adafruit |
| WR2 | 22 AWG stranded, red + black | Battery pack leads | 1 | $0.00–8.00 | $0.00–8.00 | LAB | Lab |
| **Subtotal — Interface circuits (on the Proto board)** | | | | | **$5.15–50.85** | | |
| **4 · Power** | | | | | | | |
| BT | Saft LS33600 (solder tabs) | 3.6 V Li-$SOCl_2$ D cell, 17 Ah | 2 | $22.00–35.00 | $44.00–70.00 | LATER | BatteryGuy / DigiKey |
| D1, D2 | TI LM66100 (SC-70-6) | Ideal diode per cell | 2 | $0.60–1.20 | $1.20–2.40 | LATER | DigiKey / Mouser |

| ITEM | PART NUMBER | DESCRIPTION | QTY | UNIT | EXTENDED | STATUS | SOURCE |
|---|---|---|---|---|---|---|---|
| AD2 | SC-70-6 to DIP adapter | Lets the LM66100 go on perfboard | 2 | $1.50–4.00 | $3.00–8.00 | LATER | DigiKey / Chip Quik |
| F1 | Bourns MF-R050 PTC (0.5 A hold) † | Resettable fuse at the pack | 1 | $0.50–1.50 | $0.50–1.50 | LATER | DigiKey / Mouser |
| P1 | Adafruit 261 | JST-PH 2-pin cable (pack plug + bench lead) | 2 | $0.75 | $1.50 | ORDERED | Adafruit |
| BB | Adafruit 4654 | MiniBoost 5 V (backup only) | 1 | $3.95 | $3.95 | ORDERED | Adafruit |
| HZ | Hazmat shipping | Lithium-metal cells ship ground only | 1 | $10.00–30.00 | $10.00–30.00 | LATER | Seller |
| **Subtotal — Power** | | | | | **$64.15–117.35** | | |
| **5 · RF / antenna** | | | | | | | |
| A1 | Adafruit 4269 | Spring antenna 915 MHz (buoy + shore bench) | 2 | $0.95 | $1.90 | ORDERED | Adafruit |
| A2 | Adafruit 1661 | uFL SMT connector (on the M0) | 1 | $0.75 | $0.75 | ORDERED | Adafruit |
| A3 | Adafruit 851 | SMA (panel) to uFL cable | 1 | $3.95 | $3.95 | ORDERED | Adafruit |
| A4 | 915 MHz whip, SMA male, ≤ 2.15 dBi | Deployed antenna on the lid | 1 | $10.00–15.00 | $10.00–15.00 | LATER | DigiKey / Mouser |
| **Subtotal — RF / antenna** | | | | | **$16.60–21.60** | | |
| **6 · Shore station** | | | | | | | |
| SS1 | Adafruit 5477 | ESP32-S3 Feather 4 MB / 2 MB PSRAM | 1 | $17.50 | $17.50 | ORDERED | Adafruit |
| SS2 | Adafruit 3231 | LoRa Radio FeatherWing RFM95W 900 MHz | 1 | $19.95 | $19.95 | ORDERED | Adafruit |
| SS3 | Adafruit 2830 | Stacking headers (shore stack) | 1 | $1.25 | $1.25 | ORDERED | Adafruit |
| SS4 | USB-C data cable + 5 V adapter | Shore power | 1 | $0.00–10.00 | $0.00–10.00 | LAB | Team / lab |
| **Subtotal — Shore station** | | | | | **$38.70–48.70** | | |
| **7 · Housing electrical consumables** | | | | | | | |

| ITEM | PART NUMBER | DESCRIPTION | QTY | UNIT | EXTENDED | STATUS | SOURCE |
|---|---|---|---|---|---|---|---|
| CG | Cable glands G1–G3 | From lab stock (decision 2026-09-24) | 3 | $0.00 | $0.00 | LAB | Lab |
| MS | industrial microSD 16–32 GB | Deployment logging incl. audio | 1 | $12.00–18.00 | $12.00–18.00 | LATER | DigiKey / Mouser |
| CO | Silicone conformal coating | Protects boards from condensation | 1 | $15.00–22.00 | $15.00–22.00 | LATER | DigiKey / Amazon |
| DS | Silica-gel desiccant packs | Keeps the housing dry | 1 | $5.00–8.00 | $5.00–8.00 | LATER | Amazon |
| **Subtotal — Housing electrical consumables** | | | | | **$32.00–48.00** | | |
| **8 · Optional (recommended)** | | | | | | | |
| OP1 | Adafruit 3178 | Second Feather M0 so firmware and hardware work in parallel | 1 | $34.95 | $34.95 | OPTIONAL | Adafruit |
| OP2 | Spares kit | DS18B20, turbidity sensor, microSD, antenna | 1 | $60.00–70.00 | $60.00–70.00 | OPTIONAL | Adafruit / Mouser |
| **Subtotal — Optional (recommended)** | | | | | **$94.95–104.95** | | |
| **9 · Shipping and tax already paid** | | | | | | | |
| SH1 | Invoice 3758294 | Adafruit shipping $7.70 + tax $11.98 | 1 | $19.68 | $19.68 | ORDERED | Adafruit |
| **Subtotal — Shipping and tax already paid** | | | | | **$19.68** | | |
| **Spent so far (Adafruit, incl. shipping + tax)** | | | | | **$142.58** | | |
| **To place now (Mouser)** | | | | | **$28.30** | | |
| **Planned follow-up orders** | | | | | **$331.05–425.45** | | |
| **TOTAL — prototype electronics (required)** | | | | | **$502.43–652.33** | | |
| **TOTAL — including optional items** | | | | | **$597.38–757.28** | | |

Status: ORDERED on invoice 3758294 · MOUSER order to place now · LATER planned follow-up order · LAB from lab stock · OPTIONAL not required for the prototype.

# Sheet 5 — Equipment takeoff



Tools and test equipment needed to build and verify the electronics. "Lab" items are borrowed from rooms 4017–4019; the "to buy" total counts only what the lab or the team doesn't already have.

| ITEM | WHAT IT'S FOR | QTY | EST. COST | SOURCE |
|---|---|---|---|---|
| **A · Soldering and assembly** | | | | |
| Temperature-controlled soldering station | Headers, SOT-23 adapters, uFL connector | 1 | $0 (borrow) | Lab |
| Fine conical / 1 mm chisel tips | SOT-23 MOSFETs and the uFL pads | 1 | $0.00–8.00 | Lab or buy |
| Rosin-core solder, 0.5–0.8 mm | All joints | 1 | $8.00–12.00 | Buy |
| Flux pen | Small SMD joints (SOT-23, uFL) | 1 | $6.00–9.00 | Buy |
| Desoldering braid | Removing the MCP73831 charger, fixing bridges | 1 | $4.00–6.00 | Buy |
| ESD-safe fine tweezers | Placing SOT-23 parts | 1 | $5.00–8.00 | Buy |
| Helping hands / PCB holder | Holding boards while soldering | 1 | $0.00–15.00 | Lab or buy |
| Flush cutters + wire strippers | Leads, jumper wires | 1 | $0.00–18.00 | Lab or buy |
| Heat-shrink assortment | Insulating battery and sensor leads | 1 | $6.00–9.00 | Buy |
| Hot-air / heat gun | Shrinking heat-shrink, removing the charger chip | 1 | $0 (borrow) | Lab |
| Fume extractor | Soldering safety | 1 | $0 (borrow) | Lab |
| **Subtotal to buy — Soldering and assembly** | | | **$29.00–85.00** | |
| **B · Test and measurement** | | | | |
| Digital multimeter with µA range | Sleep current, voltages, continuity | 1 | $0 (borrow) | Lab |
| Current-limited bench power supply | 3.6 V stand-in for the D cells | 1 | $0 (borrow) | Lab |
| Oscilloscope | Inrush, rail dips, preamp signal | 1 | $0 (borrow) | Lab |
| Power profiler (e.g. Nordic PPK2) | Logs µA sleep current over a whole wake cycle (optional, strongly recommended) | 1 | $0.00–100.00 | Optional buy |
| Reference thermometer | Checks DS18B20 ±0.5 °C | 1 | $10.00–20.00 | Buy |

| ITEM | WHAT IT'S FOR | QTY | EST. COST | SOURCE |
|---|---|---|---|---|
| Micro-USB + USB-C data cables | Programming the M0 and the ESP32-S3 | 2 | $0.00–20.00 | Have / buy |
| Laptop with Arduino IDE / PlatformIO | Firmware upload, serial monitor | 1 | $0 (borrow) | Have |
| **Subtotal to buy — Test and measurement** | | | **$10.00–140.00** | |
| **C · Prototyping** | | | | |
| Full-size breadboards | Bench builds, buoy and shore | 2 | $0.00–20.00 | Have / lab |
| Jumper wire kit + alligator clips | Bench wiring and probing | 1 | $0.00–10.00 | Have / lab |
| **Subtotal to buy — Prototyping** | | | **$0.00–30.00** | |
| **D · Water testing** | | | | |
| Test tank or bucket (≈ 20 L) | Sensor tests in water | 1 | $0 (borrow) | Lab fish tank |
| Sea salt + hydrometer | Seawater at the right salinity | 1 | $10.00–18.00 | Buy |
| Kaolin clay (or clean soil) | Clear vs turbid water check | 1 | $0.00–8.00 | Buy / free |
| Ice + warm water | Temperature step tests | 1 | $0 (borrow) | Have |
| **Subtotal to buy — Water testing** | | | **$10.00–26.00** | |
| **E · Safety** | | | | |
| Safety glasses | Soldering, lithium cells | 1 | $0.00–8.00 | Lab or buy |
| Fire-resistant battery bag | Storing the lithium D cells | 1 | $10.00–15.00 | Buy |
| ESD wrist strap + mat | Protecting the boards | 1 | $0.00–20.00 | Lab or buy |
| **Subtotal to buy — Safety** | | | **$10.00–43.00** | |
| **TOTAL equipment to buy (if the lab or team doesn't have it)** | | | **$59.00–324.00** | |

# Sheet 6 — General notes, checks and references



## General notes

1. All logic and sensors run at 3.3 V from the Feather's AP2112K regulator. No 5 V rail in the default build (MiniBoost 4654 is a backup only).
2. Remove the Feather's MCP73831 charger before any primary lithium cell is connected. Bench testing uses a current-limited 3.6 V supply instead of the cells.
3. Batteries connect in parallel through one ideal diode each, then the PTC fuse, then the keyed JST-PH plug.
4. Turbidity board and hydrophone circuit are powered only while sampling (Q1 on D11, Q2 on A3, LOW = on; 100 kΩ pull-ups keep them off during sleep and boot).
5. Hydrophone bias: 3.3 V through R9 100 Ω + R4 1.1 kΩ gives ≈ 2.46 V at the plug at 0.7 mA (minimum 2 V), matching Aquarian's test condition (2.5 V compliance, 1.1 kΩ).
6. Expected hydrophone level: −172 dB re 1 V/µPa at a 120 dB re 1 µPa reef soundscape gives ≈ 2.5 mV; preamp gain ≈ 230 brings it to ≈ 0.6 V for the 3.3 V ADC †.
7. Audio is stored on the SD card only; LoRa carries summary numbers (one 1-minute clip would need hours of airtime).
8. Never transmit without an antenna attached. Deployed antenna gain must not exceed 2.15 dBi (RFM95 FCC grant).
9. I²C addresses are unique (0x44 SHT40, 0x68 PCF8523). SPI chip selects: D8 radio, D10 SD; only one LOW at a time.
10. Firmware polarity: config.h says the sensor gate is HIGH = on, but the P-FET switches here are LOW = on. Firmware must drive D11 and A3 LOW to power the sensors.
11. DS18B20 data stays on D12 (firmware). The Rev A schematic label D5 is outdated; D5 stays the PCF8523 interrupt in firmware. D6 is left free in case the radio IO1 pad is ever jumpered for LoRaWAN.

## Checked against the sources

✓ H2dM: −172 dB, 1.1 kΩ output, 0.7 mA, min PIP 2 V, 3.5 mm TRS dual-mono (Aquarian H2dM manual p. 1–2)

✓ H2dM price $229 base; 3 / 6 / 9 / 15 m cables (aquarianaudio.com H2d page)

✓ DS18B20: 3.0–5.5 V, 4.7 kΩ pull-up, 750 ms (Maxim datasheet p. 6, 8, 19)

✓ Grove turbidity: 3.3 V / 5 V, output falls as turbidity rises (Seeed wiki)

✓ SHT40: I²C 0x44, 10 kΩ pull-ups on board (Adafruit guide)

✓ Adalogger: RTC on I²C, SD CS = GPIO 10 on M0 (Adafruit guide p. 6–7)

✓ Feather M0: BAT divider 100 k/100 k to D9 = AIN7; radio D3/D4/D8 (schematic rev B)

✓ DMG2305UX: P-channel, −20 V, $R_{DS(on)}$ ≤ 100 mΩ at −2.5 V, $V_{GS(th)}$ −0.5 to −0.9 V, $V_{GS}$ ±8 V max (Diodes DS36196 p. 1–2, saved in DATASHEETS)

✓ Adalogger schematic: SD CS on GPIO10 with 100 k pull-up, 10 k I²C pull-ups, RTC INT1 on an unconnected pad (DATASHEETS/feather_schem.png)

✓ Adafruit 851 = uFL to panel-mount SMA (not RP-SMA), 15 cm RG178; 1661 = uFL SMT connector, $0.75 (adafruit.com)

✓ A1–A5 work as digital I/O; radio uses #8/#3/#4 internally (Adafruit Feather M0 LoRa pinout page)

✓ Adafruit prices from invoice 3758294 and adafruit.com (2026-10-06)

## Not yet verified (marked †)

Preamp resistor and capacitor values, PTC rating, ideal-diode part, stack height, Grove current draw, Saft cell price at the chosen seller, whip antenna part number.

## References

Aquarian Audio H2d hydrophone page and H2dM user's guide (saved in DATASHEETS) · Maxim DS18B20 datasheet rev 042208 · Seeed Grove turbidity wiki · Adafruit SHT40 guide · Adafruit Adalogger FeatherWing guide · Adafruit Feather M0 RFMxx schematic rev B · Semtech SX1276 datasheet · S.C.O.U.T. Power Budget Update 2026-10-05 and Electronics Build and Test Plan.

# Sheet 7 — Every connection: parts, values, wires and how to join them



One table per circuit. Every resistor, capacitor, wire and joint on the build is listed here with its reference (R1, C1 …), value, both ends, and how it is joined. Build each circuit on a breadboard first. Solder it onto the FeatherWing Proto only after it passes its stage in the Electronics Build and Test Plan.

Joining methods: **Solder** means soldered on the Proto board. **Terminal** means the screw terminal (no solder on the probe wire). **Plug** means a connector. **HS** means slide heat-shrink over before soldering, then shrink it.

## 7A · DS18B20 temperature probe → D12 (Stage 1)

| REF | PART / VALUE | FROM | TO | HOW |
|---|---|---|---|---|
| W1-red | Probe VDD wire | Probe | T1 pin 1 → 3V rail | Terminal |
| W1-yel | Probe data wire (yellow or white, depending on batch) | Probe | T1 pin 2 → **D12** | Terminal |
| W1-blk | Probe GND wire (black or blue, depending on batch) | Probe | T1 pin 3 → GND rail | Terminal |
| R1 | 4.7 kΩ (comes in the probe bag) | T1 pin 2 (data) | 3V rail | Solder |
| T1 | 3-pin screw terminal, 3.5 or 5 mm pitch | Proto board edge | | Solder |

Prep: thread the cable through gland G1 first. Strip 5 mm off each wire, twist the strands, and tin them only lightly so the screw still bites. Check with a meter: about 3.3 V between red and black, and about 3.3 V on data at idle because of the pull-up.

## 7B · Turbidity sensor and its power switch Q1 → A1, D11 (Stage 1 direct, Stage 3 switched)

| REF | PART / VALUE | FROM | TO | HOW |
|---|---|---|---|---|
| Q1 | DMG2305UX P-MOSFET on SOT-23 adapter | S = 3V rail · G = **D11** | D = TURB_3V3 node | Solder (MOSFET to adapter first) |
| R2 | 100 kΩ | Q1 gate | 3V rail (keeps it OFF) | Solder |
| C1 | 100 µF 10 V electrolytic | TURB_3V3 (+ lead) | GND (– stripe) | Solder, watch polarity |
| C2 | 0.1 µF ceramic | TURB_3V3 | GND, next to Q1 | Solder |
| W3-red | Grove cable VCC | Grove board | TURB_3V3 | Solder pin to Proto pad |
| W3-blk | Grove cable GND | Grove board | GND rail | Solder |

| REF | PART / VALUE | FROM | TO | HOW |
|---|---|---|---|---|
| W3-yel | Grove cable SIG | Grove board | **A1** (no divider) | Solder |
| W3-wht | Grove cable NC | Not used: cut it short and insulate the end | | HS |

Set the switch on the Grove board to **A** (analog). Firmware: D11 LOW = on (SCO-186). At turn-on, check the 3V rail on the scope. If it dips more than about 100 mV, change C1 to 10 µF †.

### 7C · SHT40 humidity / leak sensor → I²C (Stage 1)

| REF | PART | WIRE COLOUR → PROTO PAD | HOW |
|---|---|---|---|
| W5 | STEMMA QT → male cable (Adafruit 4209) | red → 3V · black → GND · blue → SDA · yellow → SCL | Plug on the sensor end, solder the pins on the Proto end |

No extra parts needed: the SHT40 and the Adalogger already have their I²C pull-ups. Mount the SHT40 high in the housing, away from the batteries.

### 7D · Hydrophone front-end → A2, A3 (Stage 7)

| REF | PART / VALUE | FROM | TO | PURPOSE |
|---|---|---|---|---|
| Q2 | DMG2305UX on SOT-23 adapter | S = 3V rail · G = **A3** | D = AUD_3V3 | Audio power only while recording |
| R3 | 100 kΩ | Q2 gate | 3V rail | Keeps it OFF while asleep |
| C8, C9 | 0.1 µF ceramic + 10 µF | AUD_3V3 | GND | Rail decoupling, C8 at U1 pin 8 |
| R9 | 100 Ω | AUD_3V3 | BIAS_F node | Filters rail noise out of the bias |
| C10 | 10 µF | BIAS_F | GND | Filters rail noise out of the bias |
| R4 | 1.1 kΩ | BIAS_F | J1 **L** (= tip) | Plug-in power: tip sits at ≈ 2.46 V, above the 2 V minimum |
| J1 | 3.5 mm jack breakout (Adafruit 1699) | L pin = tip (signal) · GND pin = sleeve | R pin (ring): leave unconnected | Hydrophone plugs in here |
| C3 | 1 µF ceramic (non-polar) | J1 L (tip) | U1 pin 3 (IN A+) | Blocks the 2.46 V DC, passes the sound |
| R5, R6 | 10 kΩ + 10 kΩ | AUD_3V3 → VMID | VMID → GND | Makes 1.65 V mid-rail |
| C4 | 10 µF | VMID | GND | Keeps the mid-rail quiet |
| R7 | 100 kΩ | VMID | U1 pin 3 | Biases stage 1 at 1.65 V |
| R10 | 20 kΩ | U1 pin 1 (OUT A) | U1 pin 2 (IN A–) | Stage 1 gain = 1 + 20k/1k = 21 |

| REF | PART / VALUE | FROM | TO | PURPOSE |
|---|---|---|---|---|
| R11 + C5 | 1 kΩ in series with 10 µF | U1 pin 2 | GND | Sets the AC gain; DC gain stays 1. Low cut ≈ 16 Hz |
| R8 | 1.5 kΩ | U1 pin 1 | U1 pin 5 (IN B+) | Low-pass filter ≈ 10.6 kHz (anti-alias) |
| C6 | 10 nF ceramic | U1 pin 5 | GND | Low-pass filter ≈ 10.6 kHz (anti-alias) |
| R12 | 10 kΩ | U1 pin 7 (OUT B) | U1 pin 6 (IN B–) | Stage 2 gain = 1 + 10k/1k = 11 |
| R13 + C7 | 1 kΩ in series with 10 µF | U1 pin 6 | GND | Stage 2 AC gain; DC stays at 1.65 V |
| U1 | MCP6002 DIP-8 in a socket | pin 8 → AUD_3V3 · pin 4 → GND | pin 7 → **A2** | Total gain ≈ 231 (47 dB) |

The electrolytic 10 µF caps C4, C5 and C7 sit at 1.65 V, so their + lead goes toward the op-amp/VMID side and the – stripe to GND. 10 µF ceramics avoid the polarity question. Bench check: with the hydrophone in a bucket and tapped, A2 should swing around 1.65 V without clipping at 0 V or 3.3 V. If it clips, lower R10 to 10 kΩ †.

### 7E · Battery pack ⟶ Feather BAT (Stage 6; remove the Feather's charger first)

| REF | PART / VALUE | FROM | TO | HOW |
|---|---|---|---|---|
| BT1, BT2 | Saft LS33600 with solder tabs | + tab | D1 / D2 VIN | 22 AWG stranded red, solder to the **tab only**, HS |
| BT1, BT2 | Cell – tabs | – tab | Pack GND bus | 22 AWG stranded black, HS |
| D1, D2 | LM66100 ideal diode on SC-70-6 adapter † | VIN ← cell + · GND → pack – · CE → GND (enabled, check datasheet) † | VOUT → common + node | Solder |
| F1 | PTC resettable fuse, 0.5 A hold (e.g. Bourns MF-R050) † | Common + node | P1 red | Solder inline, HS |
| P1 | JST-PH 2-pin cable with tinned ends (Adafruit 261) | red ← F1 · black ← pack – | Plugs into the Feather's battery jack | Solder + HS, then plug |

Never solder to the cell body and never short a cell. Before plugging in, measure P1: red should read +3.5 to 3.7 V relative to black. Cable-tie and foam-pad both cells in the bay, and store spares in the fire bag.

### 7F · Antenna and board stack (Stages 1, 4, 8)

| REF | PART | WHAT TO DO |
|---|---|---|
| H1, H2 | Stacking headers 2830 (×2) | Solder onto the M0 and the Adalogger, sockets facing up. Keep the pins straight by plugging them into a breadboard while soldering. |
| H3 | Male headers, 12 + 16 pin | Solder on the bottom of the Proto board so it plugs into the Adalogger sockets. Use the board's own headers or lab stock. |
| CR1 | CR1220 coin cell | Into the Adalogger holder, + side up. |
| ANT-b | Spring antenna 4269 (bench) | Solder to the M0 ANT pad. Remove it before fitting the uFL path. |
| ANT-d | uFL 1661 → cable 851 → whip | Solder 1661 to the M0 uFL footprint (fine tip, flux). Snap on 851, put its SMA through the lid and tighten the nut, then screw on the whip. |
| SS | Shore: ESP32-S3 + LoRa wing | Stacking headers on the ESP32-S3. On the wing, bridge the CS, RST and IRQ jumpers to the pins in the firmware, and solder the spring antenna. |

## Build order (follows the Electronics Build and Test Plan)

1. **Stage 1:** headers on the M0 and the Adalogger, coin cell in. On a breadboard: DS18B20 + R1, SHT40 via 4209, turbidity straight to 3V and A1. Get first readings over USB.
2. **Stage 2:** measure sleep and active current with the meter in series.
3. **Stage 3:** build Q1 + R2 + C1 + C2 on the breadboard and check that D11 LOW turns the sensor on. Then remove the Feather's charger.
4. **Stage 4:** spring antennas on both radios; shore jumpers bridged. Range test.
5. **Stage 6:** build the battery pack (7E), check its polarity, then the 7-day run.
6. **Stage 7:** breadboard the hydrophone front-end (7D), check A2, record test clips.
7. **Stage 8:** move all circuits onto the Proto board, terminals and jack at the edges. Thread cables through the glands before terminating. Fit the uFL → SMA path. Conformal-coat the boards (keep coating off the connectors and the SD slot), add desiccant, seal.